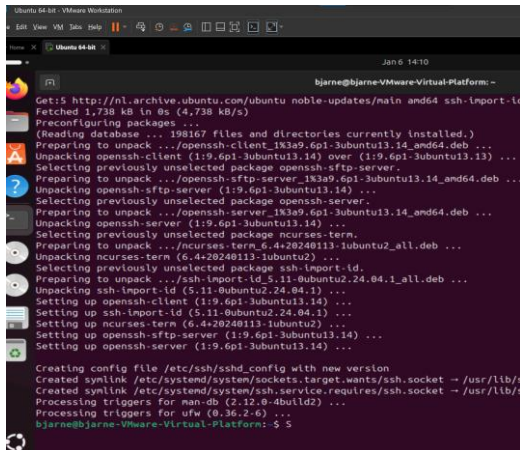


Template Week 6 – Networking

Student number: 587890

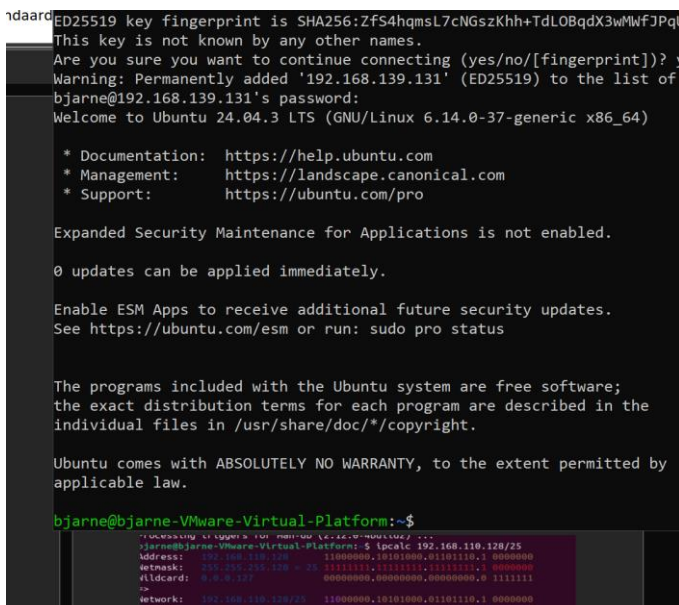
Assignment 6.1: Working from home

Screenshot installation openssh-server:



```
Get:5 http://nl.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id
Fetched 1,738 kB in 0s (4,738 kB/s)
Preconfiguring packages ...
(Reading database ... 198167 files and directories currently installed.)
Preparing to unpack .../openssh-client_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-client (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.13) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Creating config file /etc/ssh/sshd_config with new version
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/s
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/s
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
bjarne@bjarne-VMware-Virtual-Platform:~$
```

Screenshot successful SSH command execution:



```
ED25519 key fingerprint is SHA256:ZfS4hqmsL7cNGszKhh+TdLOBqdX3wMwfJPqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Warning: Permanently added '192.168.139.131' (ED25519) to the list of
bjarne@192.168.139.131's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

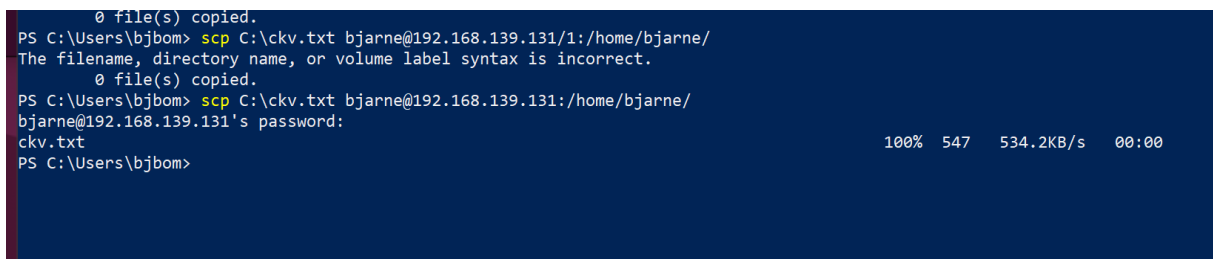
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

bjarne@bjarne-VMware-Virtual-Platform:~$
```

Screenshot successful execution SCP command:



```
0 file(s) copied.
PS C:\Users\bjbom> scp C:\ckv.txt bjarne@192.168.139.131:/home/bjarne/
The filename, directory name, or volume label syntax is incorrect.
0 file(s) copied.
PS C:\Users\bjbom> scp C:\ckv.txt bjarne@192.168.139.131:/home/bjarne/
bjarne@192.168.139.131's password:
ckv.txt
PS C:\Users\bjbom>
100% 547 534.2KB/s 00:00
```

Screenshot remmina:

Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:

```
C:\Windows\system32\cmd.exe - nslookup
Name: amazon.com
Addresses: 98.87.179.71
          98.82.161.185
          98.87.179.74

> google.com
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
Non-authoritative answer:
Name: google.com
Addresses: 2001:4860:4000::200e
          142.250.179.142

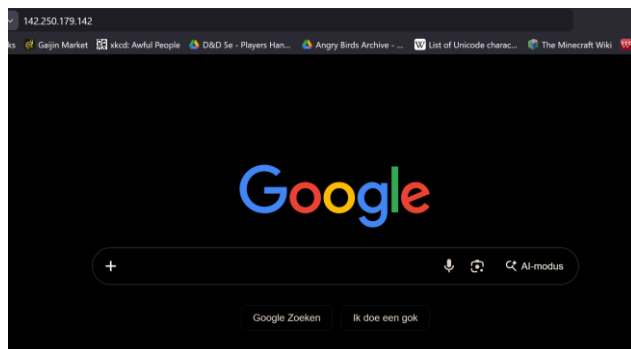
> mome.one.one.one
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
Name: mome.one.one.one

> nads.google.com
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
*** d-hk-mer-1b02.infra.saxion.net can't find no
> dns.google.com
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
Non-authoritative answer:
Name: dns.google.com
Addresses: 2001:4860:4000::8844
          2001:4860:4000::8888
          8.8.4.4
          8.8.8.8

> bnl.com
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
Non-authoritative answer:
Name: bnl.com
Address: 79.170.100.62

> wiscnools.com
Server: d-hk-mer-1b02.infra.saxion.net
Address: 145.2.14.10
Non-authoritative answer:
Name: wiscnools.com
Addresses: 76.223.115.82
          11.246.240.135
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

$$32 - 25 = 7$$

$$2^7 = 128$$

What is the usable IP range to hand out to the connected computers?

$$128 - 2 = 126$$

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

```
Processing triggers for Man-db (2.12.0-4ubuntu2) ...
bjarne@bjarne-VMware-Virtual-Platform:~$ ipcalc 192.168.110.128/25
Address:    192.168.110.128      11000000.10101000.01101110.1 00000000
Netmask:    255.255.255.128 = 25 11111111.11111111.11111111.1 00000000
Wildcard:    0.0.0.127          00000000.00000000.00000000.0 11111111
=>
Network:    192.168.110.128/25  11000000.10101000.01101110.1 00000000
HostMin:    192.168.110.129     11000000.10101000.01101110.1 00000001
HostMax:    192.168.110.254     11000000.10101000.01101110.1 11111110
Broadcast:  192.168.110.255     11000000.10101000.01101110.1 11111111
Hosts/Net:  126                  Class C, Private Internet

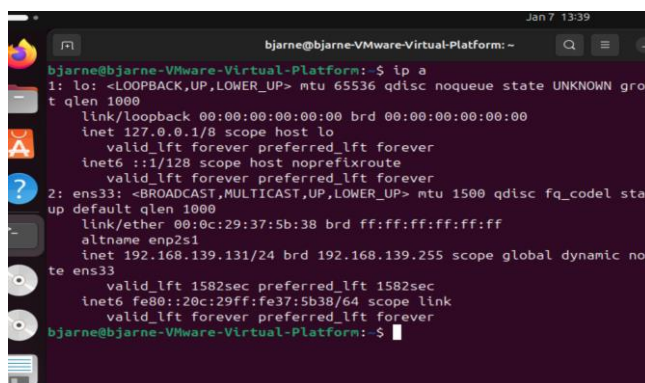
bjarne@bjarne-VMware-Virtual-Platform:~$ S
```

Explain the above calculation in your own words.

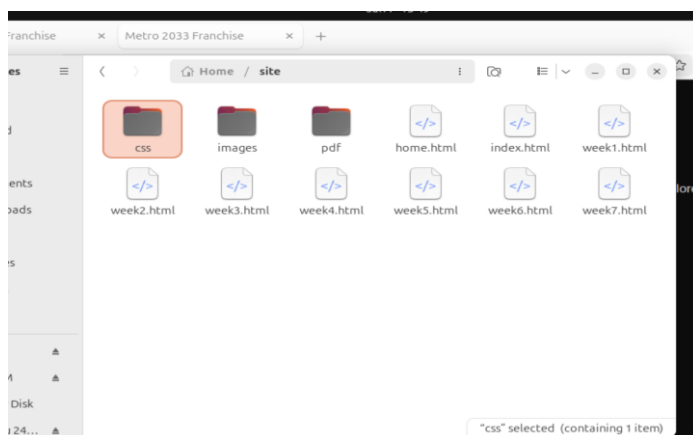
The first and last ip address is network address and broadcast address. So can not be given out to users

Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:



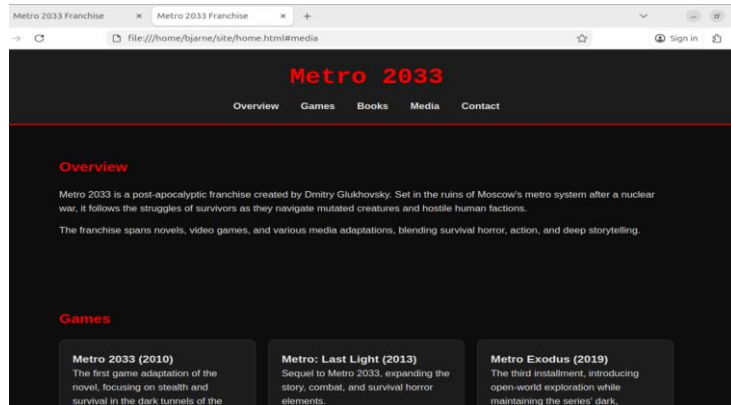
Screenshot of Site directory contents:



Screenshot python3 webserver command:

```
valid_lft forever preferred_lft forever
bjarne@bjarne-VMware-Virtual-Platform:~$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
127.0.0.1 - - [07/Jan/2026 13:51:46] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [07/Jan/2026 13:51:46] code 404, message File not found
127.0.0.1 - - [07/Jan/2026 13:51:46] "GET /favicon.ico HTTP/1.1" 404 -
S
```

Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

Example: 192.168.1.100/27

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100

Subnet Mask: 11111111.11111111.11111111.11100000

Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.

For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).

The range of this network segment is from 192.168.1.96 to 192.168.1.127.

Paste source code here, with a screenshot of a working application.

```
import java.util.Scanner;
```

```
public class NetworkCalculator {
```

```

// Convert a decimal IP part to an 8-bit binary string
public static String toBinaryString(int num) {
    String binary = Integer.toBinaryString(num);
    while (binary.length() < 8) {
        binary = "0" + binary;
    }
    return binary;
}

// Convert an IP string "192.168.1.100" to an int array [192,168,1,100]
public static int[] parseIP(String ip) {
    String[] parts = ip.split("\\.");
    int[] ipNums = new int[4];
    for (int i = 0; i < 4; i++) {
        ipNums[i] = Integer.parseInt(parts[i]);
    }
    return ipNums;
}

// Convert int array to string IP
public static String toDecimalIP(int[] ip) {
    return ip[0] + "." + ip[1] + "." + ip[2] + "." + ip[3];
}

// Convert int array to binary string IP
public static String toBinaryIP(int[] ip) {
    return toBinaryString(ip[0]) + "." + toBinaryString(ip[1]) + "."
        + toBinaryString(ip[2]) + "." + toBinaryString(ip[3]);
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    // Input IP address
    System.out.print("Enter IP address (e.g., 192.168.1.100): ");
    String ipInput = scanner.nextLine();
    int[] ip = parseIP(ipInput);

    // Input subnet mask
    System.out.print("Enter subnet mask (e.g., 255.255.255.224): ");
    String subnetInput = scanner.nextLine();
    int[] subnet = parseIP(subnetInput);

    // Calculate network address using bitwise AND
    int[] network = new int[4];
    for (int i = 0; i < 4; i++) {
        network[i] = ip[i] & subnet[i];
    }
}

```

```

// Calculate the number of addresses in the subnet
int hostBits = 0;
for (int i = 0; i < 4; i++) {
    int mask = subnet[i];
    while ((mask & 1) == 0) {
        hostBits++;
        mask >>= 1;
    }
}
int numAddresses = (int) Math.pow(2, hostBits);

// Calculate broadcast address
int[] broadcast = new int[4];
for (int i = 0; i < 4; i++) {
    broadcast[i] = network[i] | (~subnet[i] & 0xFF);
}

// Output results
System.out.println("\nIP Address (binary): " + toBinaryIP(ip));
System.out.println("Subnet Mask (binary): " + toBinaryIP(subnet));
System.out.println("-----");
System.out.println("Network Address (binary): " + toBinaryIP(network));
System.out.println("Network Address (decimal): " + toDecimalIP(network));
System.out.println("Broadcast Address: " + toDecimalIP(broadcast));
System.out.println("Number of IPs in subnet: " + numAddresses);
System.out.println("IP Range: " + toDecimalIP(network) + " - " + toDecimalIP(broadcast));
}
}

```

```

NetworkCalculator x
C:\Users\bjbom\.jdk\ms-21.0.8\bin\java.exe "-javaagent:C:\Program Fil
Enter IP address (e.g., 192.168.1.100): 192.168.1.100
Enter subnet mask (e.g., 255.255.255.224): 255.255.255.224

IP Address (binary):      11000000.10101000.00000001.01100100
Subnet Mask (binary):    11111111.11111111.11111111.11100000
-----
Network Address (binary): 11000000.10101000.00000001.01100000
Network Address (decimal): 192.168.1.96
Broadcast Address:       192.168.1.127
Number of IPs in subnet: 32
IP Range: 192.168.1.96 - 192.168.1.127

Process finished with exit code 0

```

Ready? Save this file and export it as a pdf file with the name: [week6.pdf](#)